

TP2 filtering

Exercise 1:

Generate a sine wave (amplitude $A=1$, normalized frequency $f_0=0.09814$, $nbs=32768$ samples)

add a white Gaussian noise (`np.random.randn(nbs)`) so that the SNR will be 20 dB.

Plot the temporal view

Plot the psd over $M=1024$ frequencies.

try different values of $M=(32768, 1024, 256)$

your conclusions?

Exercise 2:

Load the the signal 1 : `sig1=np.loadtxt("sig1.txt")`

analysis this signal

Assuming that the sampling frequency is 100 kHz. Identify the true frequencies of all the signals.

Exercise 3:

Load the the signal 2 :

`samplerate, sig2 = wavfile.read('signal.wav')` (rajouter la librairie from `scipy.io import wavfile`)

`print(sig2.shape)`

`print(samplerate)`

analysis this signal